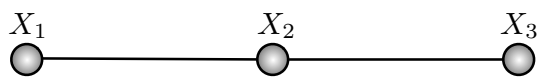
Statistical Models

TUHH, Summer Term 2024

Exercise Sheet 4 (Ay, July 11, 2024)

Exercise 4.1 Let V be a non-empty finite set, and let A be a non-empty subset of V. Prove that the boundary of A, $\mathrm{bd}(A)$, separates A from the complement of its closure in V, that is $V \setminus \mathrm{cl}(A)$.

Exercise 4.2 The following figure shows a graph G with node set $V = \{1, 2, 3\}$ and associated variables X_1, X_2, X_3 which we assume to be binary, say with values in $\{0, 1\}$.



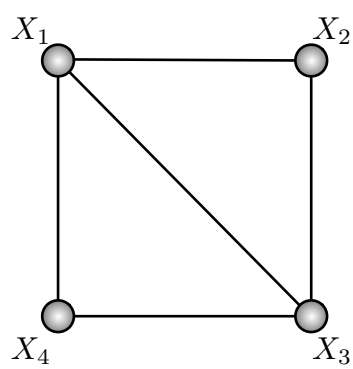
Furthermore, we assume that the joint distribution of X_1, X_2, X_3 has the following structure associated with G:

$$p(x_1, x_2, x_3) \;=\; \frac{1}{Z(\theta)} \exp\left(\theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \theta_{12} x_1 x_2 + \theta_{23} x_2 x_3\right),$$

where $\theta = (\theta_1, \theta_2, \theta_3, \theta_{12}, \theta_{23}) \in \mathbb{R}^5$. Show that the following holds:

$$p(x_1, x_3 | x_2) \;=\; p(x_1|x_2)\, p(x_3|x_2).$$

Exercise 4.3 The following figure shows a graph G with node set $V = \{1, 2, 3, 4\}$ and associated variables X_1, X_2, X_3, X_4 which we assume to be binary, say with values in $\{0, 1\}$.



1. Determine the dimension of the simplex of *all* positive probability measures on $\{0,1\}^4$.

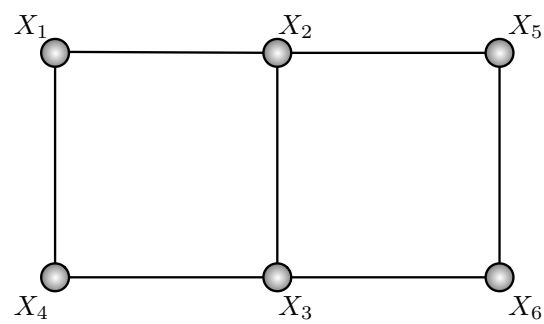
2. List all the non-empty cliques of G.

3. Determine the dimension of the graphical model associated with G, that is the exponential family of positive probability measures on $\{0,1\}^4$ that factorise according to G.

4. Determine the dimension of the independence model, that is the exponential family of positive product probability measures on $\{0,1\}^4$.

5. Show that the independence model is a sub-model of the graphical model associated with G.

Exercise 4.4 The following figure shows a graph G with node set $V = \{1,2,3,4,5,6\}$ and associated variables $X_1, X_2, X_3, X_4, X_5, X_6$ with a distribution that factorises according to G.



1. List all local Markov properties with respect to G.

2. List all pairwise Markov properties with respect to G.

3. List at least five global Markov properties that are different from the local and pairwise Markov properties listed in 1. and 2.

Exercise 4.5 Compute the stationary probability measure of the following stochastic matrix:

$$P = \frac{1}{12}\begin{pmatrix} 2 & 5 & 1 & 4 \\ 5 & 1 & 4 & 2 \\ 1 & 4 & 2 & 5 \\ 4 & 2 & 5 & 1 \end{pmatrix}$$